

ADVANCED DATA MANAGEMENT 2023/2024
January 9, 2025
CLOSED BOOK

FIRST NAME:
SID:

LAST NAME:

Q1. (up to 10 lines) [**only for COMPUTER ENGINEERING STUDENTS**]
Explain the CAP theorem, describing the three CAP properties and the main result; briefly discuss its impact on the design of large-scale data management systems.

Q2. (up to 10 lines)
Discuss how associations can be represented at design time inside aggregation-based data schemas, presenting some examples.

Q3. (up to 10 lines)

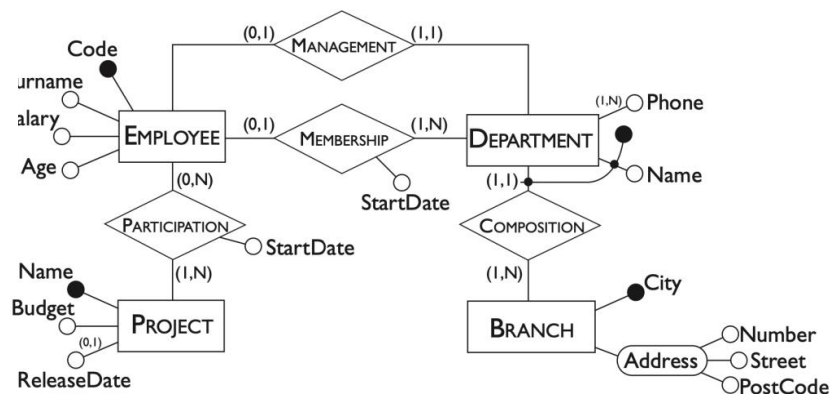
Discuss *hash-based partitioning over a ring*, explaining what the set of hash values consists of, which set of values the hash function is applied to, and how read operations are performed, considering both the best-case and worst-case scenarios

Q4. (up to 10 lines) [**only for COMPUTER SCIENCE STUDENTS**]

Describe the main differences between RDFS and OWL, presenting some examples.

Exercise 1 [NoSQL aggregate-oriented data stores]

Consider the following entity relationship diagram:



The workload is the following:

- **Query 1:** Determine the name and the release date of the projects with budget equal to 100.000 euros.
- **Query 2:** Determine the code and the surname of the employees that participate in the project named "P10".
- **Query 3:** Determine the code of the employees that are member of departments named "D5".

Based on the previous ER schema and workload:

1. Design, using the aggregate-oriented logical design methodology, the aggregates needed for answering the given workload queries. Present the output of each step of the methodology.

Queries

Q1

Q2

Q3

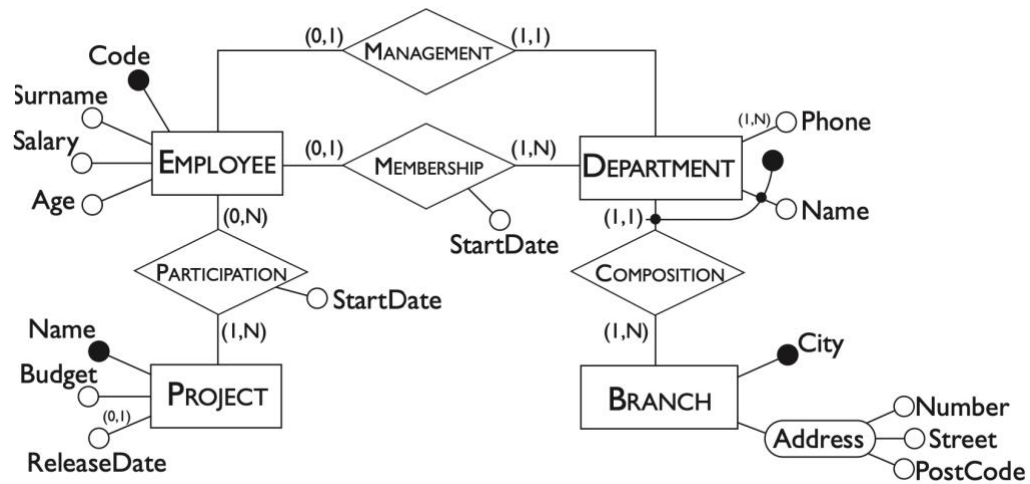
Annotation: please annotate the input ER diagram

Aggregates

2. Suppose the selected system is Cassandra. Revise each aggregate schema, in meta-notation, to obtain a set of aggregate schemas for Cassandra, specified in meta-notation, so that queries Q1, Q2, Q3 can be executed without the creation of any index and without the usage of the ALLOW FILTERING clause. Motivate your choices.
3. For each aggregate schema designed for Cassandra, choose a suitable partition key and a suitable primary key, motivating your choice.

Exercise 2 [NoSQL graph-based data stores]

Consider the entity relationship diagram presented in Exercise 1:



1. Present the most suitable logical schema in a graph-based data store, like Neo4j (in the form of a graph pattern or of a sample instance), for the domain described by the Entity Relationship diagram.
2. Formulate in Cypher the query to determine the projects in which participates only employees that are member of no department.

Exercise 3 [Semantic data stores] [only for COMPUTER SCIENCE STUDENTS]

Consider the following RDF excerpt:

```
ex:player1 rdf:type ex:Player .
ex:player1 ex:hasName "Jannik Sinner" .
ex:player1 ex:hasNationality ex:Italian .
ex:player1 ex:playsIn ex:RolandGarros2025 .
ex:player1 ex:hasRanking "1" .

ex:player2 rdf:type ex:Player .
ex:player2 ex:hasName "Carlos Alcaraz" .
ex:player2 ex:hasNationality ex:Spanish .
ex:player2 ex:playsIn ex:RolandGarros2025 .
ex:player2 ex:hasRanking "2" .

ex:player3 rdf:type ex:Player .
ex:player3 ex:hasName "Alexander Zverev" .
ex:player3 ex:hasNationality ex:German .
ex:player3 ex:playsIn ex:RolandGarros2025 .
ex:player3 ex:hasRanking "3" .

ex:player4 rdf:type ex:Player .
ex:player4 ex:hasName "Jack Draper" .
ex:player4 ex:playsIn ex:RolandGarros2025 .
ex:player4 ex:hasRanking "4" .

ex:player5 rdf:type ex:Player .
ex:player5 ex:hasName "Novak Djokovic" .
ex:player5 ex:hasNationality ex:Serbian .
ex:player5 ex:playsIn ex:RolandGarros2025 .
ex:player6 ex:hasRanking "5" .

ex:player6 rdf:type ex:Player .
ex:player6 ex:hasName "Lorenzo Musetti" .
ex:player6 ex:hasNationality ex:Italian .
ex:player6 ex:playsIn ex:RolandGarros2025 .
ex:player6 ex:hasRanking "6" .

ex:RolandGarros2025 rdf:type ex:Tournament .
ex:RolandGarros2025 ex:hasLocation ex:Paris .
ex:RolandGarros2025 ex:hasSurface ex:Clay .
```

1. Add class specifications for `ex:Player` and `ex:Tournament`.
2. Specify `ex:playsIn` as a property, defining its domain and range.
3. Specify `ex:TopPlayer` as a subclass of `ex:Player`. Is it possible, and how, to specify that a top player is a player with ranking less than or equal to 5?

4. Is it possible, and how, to specify that a tournament has only one location? It is possible, and how, to specify that a location is associated with only one tournament?

5. Write a SPARQL query to count the Italian players participating in Roland Garros 2025, with ranking lower than 4.

6. Write a SPARQL query to retrieve the name and the nationality (if available) of all the players with a ranking higher than 3 or who play in a tournament located in London.

Hint (Cheat sheet RDF/S and OWL)

- instance relation `rdf:type` (classes are instances of `rdfs:Class`)
- subclass relation `rdfs:subClassOf`
- domain of a property `rdfs:domain`
- range of a property `rdfs:range`
- subproperty `rdfs:subPropertyOf`
- functional property `owl:functionalProperty`
- inverse functional property `owl:inverseFunctionalProperty`
- class disjointness `owl:disjointWith`
- instance equality and disequality `owl:sameAs` and `owl:differentFrom`